Raisins and additional walking have distinct effects on plasma lipids and inflammatory cytokines

❖ ❖ ❖

Background Raisins are a significant source of dietary fiber and polyphenols, which may reduce cardiovascular disease (CVD) risk by affecting lipoprotein metabolism and inflammation. Walking represents a low intensity exercise intervention that may also reduce CVD risk. The purpose of this study was to determine the effects of consuming raisins, increasing steps walked, or a combination of these interventions on blood pressure, plasma lipids, glucose, insulin and inflammatory cytokines. Results Thirty-four men and postmenopausal women were matched for weight and gender and randomly assigned to consume 1 cup raisins/d (RAISIN), increase the amount of steps walked/d (WALK) or a combination of both interventions (RAISINS + WALK). The subjects completed a 2 wk run-in period, followed by a 6 wk intervention. Systolic blood pressure was reduced for all subjects ($P = 0.008$). Plasma total cholesterol was decreased by 9.4% for all subjects ($P < 0.005$), which was explained by a 13.7% reduction in plasma LDL cholesterol (LDL-C) ($P < 0.001$). Plasma triglycerides (TG) concentrations were decreased by 19.5% for WALK ($P < 0.05$ for group effect). Plasma TNF-α was decreased from 3.5 ng/L to 2.1 ng/L for RAISIN ($P < 0.025$ for time and group × time effect). All subjects had a reduction in plasma sICAM-1 ($P < 0.01$). Conclusion This research shows that simple lifestyle modifications such as adding raisins to the diet or increasing steps walked have distinct beneficial effects on CVD risk.